Bluestock Mutual Fund Analytics Capstone

An end-to-end ETL, financial performance, risk analytics, and business intelligence study of the Indian Mutual Fund Industry — commissioned by Bluestock Fintech.

DATA ANALYTICS CAPSTONE · JULY 2026

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Engagement Snapshot

Bluestock Fintech commissioned this engagement to evaluate in-house Mutual Fund Analytics capability — spanning data engineering, performance measurement, risk analytics, and BI reporting.

10

Interlinked Datasets

Fund Master, NAV History, AUM, SIP, Category Flows, Folios, Scheme Performance, Transactions, Holdings, Benchmarks

8

Performance Metrics

Daily Return, CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown, Fund Scorecard

6

Advanced Analytics

Historical VaR, CVaR, Rolling Sharpe, Investor Cohorts, SIP Continuity, Sector HHI

4

Power BI Pages

Industry Overview, Fund Performance, Investor Analytics, SIP & Market Trends

Technology stack: Python, Pandas, NumPy, Matplotlib, SQLite, SQL, Power BI, Jupyter Notebook. Report: ~30 pages, consulting-grade format.

Project Objectives

Primary Objectives

- Build a complete ETL pipeline ingesting 10 mutual fund datasets into a structured SQLite warehouse
- Compute financial metrics (CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown) benchmarked against indices
- Perform advanced risk analytics: VaR, CVaR, Rolling Sharpe, Sector HHI, SIP Continuity, Investor Cohorts
- Design a rules-based recommendation engine mapping risk appetite to ranked fund shortlists
- Build a four-page interactive Power BI dashboard for business stakeholders

Success Criterion

Every analytical output is **reproducible end-to-end** from raw datasets through documented Python scripts, SQL schema, and Power BI data model — auditable and extendable by Bluestock's internal analytics team.

Methodology & Database Design

1

Extract

Ingest 10 CSV datasets via Pandas with initial profiling

2

Clean

Forward-fill NAV gaps, standardise transaction types, filter expense ratios

3

Validate

Re-profile, confirm referential integrity across all amfi_code joins

4

Normalise

Standardise units (₹ Crore, %), control categorical values

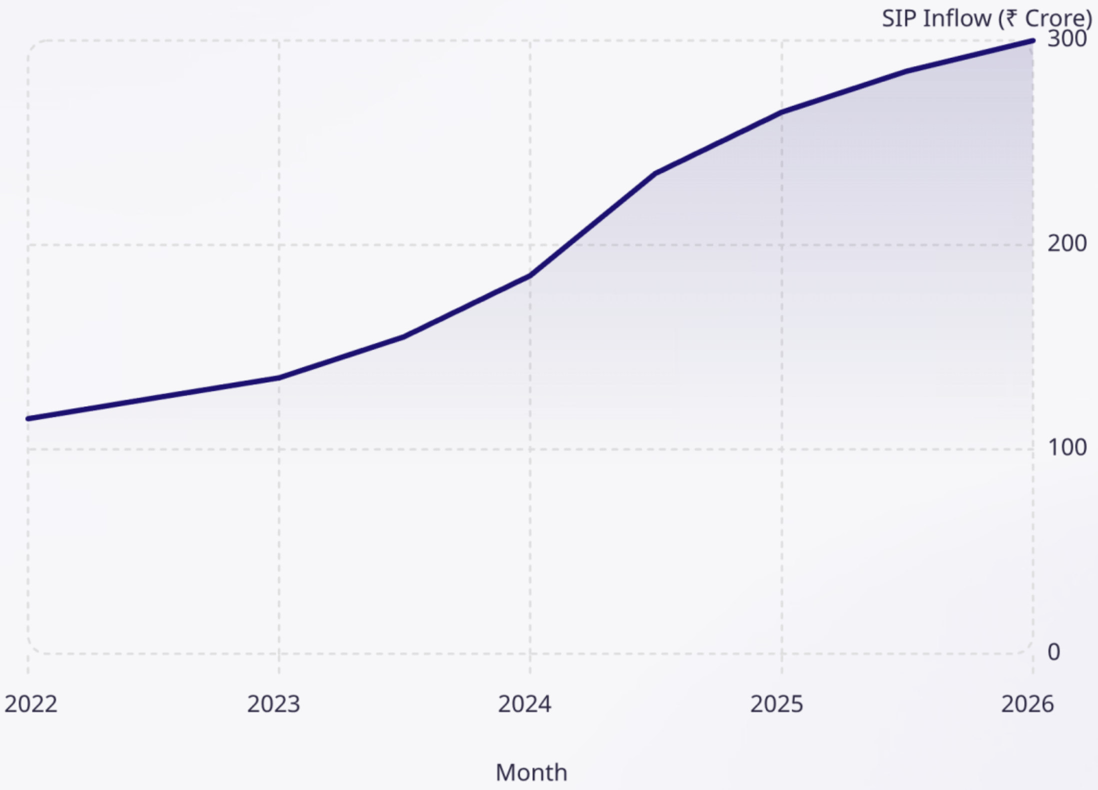
5

Load

Bulk-load into SQLite with enforced primary and foreign keys

Light star-schema in SQLite: **dim_fund** and **dim_date** surround four fact tables — **fact_nav**, **fact_transactions**, **fact_performance**, and **fact_aum** — enabling flexible scheme, fund-house, and time-based aggregation.

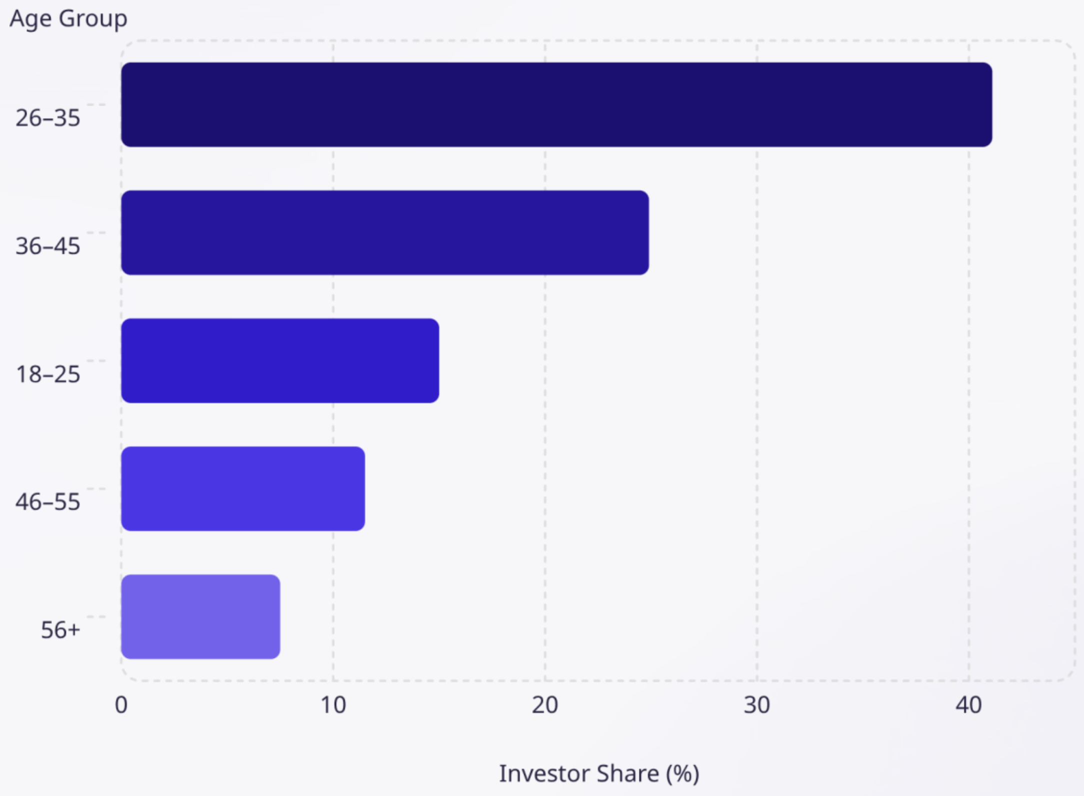
Exploratory Data Analysis — Industry Growth



Key EDA Insights

- **SIP inflows** rose from ₹115 Cr to ₹300 Cr — structurally embedded, resilient through dips
- **Industry folios** doubled from 13.2 Cr to 26.12 Cr (2022–2026)
- **Equity folios** dominate and grow fastest — growth-seeking retail base
- **AUM concentration** widening: SBI MF leads at ₹1.25 Cr, top AMCs pulling ahead
- **Category inflows** rotate monthly — dynamic, not static demand

Investor Demographics



Demographic Profile

- **66% male / 34% female** — gender gap is an addressable growth opportunity
- **66.3% metro (T30) / 33.7% Tier-2 & 3 (B30)** — geographic broadening underway
- **Younger skew** implies longer horizons, greater equity comfort
- **SIP ticket size** rises with age — tracks income stage, not risk appetite
- **Top states:** Punjab, Tamil Nadu, Madhya Pradesh lead in investment value

Performance Analytics

Top Schemes by Sharpe Ratio

Scheme	Category	Sharpe
Mirae Asset Large Cap	Equity	1.45
Kotak Flexicap	Equity	1.31
Mirae Asset Tax Saver	Equity	1.23
SBI Bluechip	Equity	1.21
ICICI Pru Midcap	Equity	1.18

Key Performance Findings

- **Highest Sharpe Ratios** in large-cap and flexi-cap — not highest-return small-caps
- **Highest Alpha** in small-cap and mid-cap (DSP Small Cap, ICICI Pru Midcap > 0.30)
- **Deepest drawdowns** exceed −50% for SBI and Axis Small Cap Funds
- **Top 10 funds** dominated by small-cap and mid-cap schemes
- **Fund Scorecard** consolidates return, Sharpe, Alpha, expense ratio, drawdown into 0–100 composite

Advanced Analytics

Historical VaR (95%)

Small-cap schemes post worst single-day tail losses (worse than −2.6% at 95% confidence)

CVaR / Expected Shortfall

Consistently worse than VaR (e.g., −3.2% vs −2.7%) — confirms meaningful additional tail severity

Rolling 90-Day Sharpe

Risk-adjusted performance compresses during volatile periods — invisible in static metrics

SIP Continuity

Substantial share of investors flagged "At Risk" with gaps beyond 30-day cadence — live retention exposure

Sector HHI

Scores range 0.12–0.25; UTI Flexi Cap most concentrated (0.25), UTI Mid Cap least (0.12)

Investor Cohorts

2024 and 2025 cohorts show comparable SIP amounts (~₹107K vs ₹109K); 2024 has higher total accumulation

Power BI Dashboard & Recommendation Engine

Four Dashboard Pages

1

Industry Overview

AUM trends, fund house comparison, SIP and folio KPIs

2

Fund Performance

Risk vs Return scatter, Fund Scorecard, benchmark comparison

3

Investor Analytics

Demographics, state-wise investors, SIP Continuity watch-list

4

SIP & Market Trends

SIP vs NIFTY overlay, NAV correlation heatmap, Rolling Sharpe

Recommendation Engine

Rules-based, interpretable engine matching investor risk appetite (Low / Moderate / High) to ranked fund shortlists.

- Filters candidate pool by risk grade
- Ranks by **Sharpe Ratio**, then 3-year return as tie-breaker
- Returns top 3 schemes with full evidence (Sharpe, return, expense ratio)
- Extensible to include expense ratio, Sector HHI, and investment horizon

Key Findings & Strategic Recommendations

Small & Mid-Cap Outperformance

Consistent risk-adjusted outperformance — but with drawdowns exceeding −50%. Always pair returns with risk context.

SIP Base: Growing but At Risk

Inflows rising, yet meaningful investor share shows lapse patterns. Build automated SIP Continuity monitoring and timed retention campaigns.

Expense Ratio ≠ Performance

Weak correlation with realised performance. Surface expense ratio prominently in all fund comparisons.

Demographic Opportunity

Younger, geographically diverse base with gender gap. Tailor onboarding by age cohort; invest in vernacular support and targeted financial literacy.

Future Scope

Cloud migration, real-time AMFI API integration, ML-based recommendations, and automated ETL refresh cycles.